

公式

2025 年 6 月 25 日

$$A_t = Q_{\pi_\theta}(s_t, a_t) - V_{\pi_\theta}(s_t) \quad (1)$$

$$a_t = [v_{xt}, v_{yt}, v_{zt}] \quad (2)$$

$$r_t^{\text{act}} = e^{-n\|d\|} \quad \text{其中} \quad d = \sqrt{(x - x_d)^2 + (y - y_d)^2 + (z - z_d)^2} \quad (3)$$

$$r_t^{\text{success}} = \begin{cases} \text{score}, & X_t \geq T \\ 0, & X_t < T \end{cases} \quad (4)$$

$$r_t^{\text{col}} = \begin{cases} \text{score}, & \text{发生碰撞} \\ r_t^{\text{act}}, & \text{其他} \end{cases} \quad (5)$$

$$r_t = w_1 r_t^{\text{act}} + w_2 r_t^{\text{col}} + w_3 r_t^{\text{success}} \quad (6)$$

$$L^{\text{clip}}(\theta) = \mathbb{E}_{s, a \sim \pi_{\theta_t}} [\min(\wp(\theta) A^{\pi_\theta}, \text{clip}(\wp(\theta), 1 - \epsilon, 1 + \epsilon) A^{\pi_{\theta_t}})] \quad (7)$$

$$\text{clip}(x, x_{\text{MIN}}, x_{\text{MAX}}) = \begin{cases} x, & x_{\text{MIN}} \leq x \leq x_{\text{MAX}} \\ x_{\text{MIN}}, & x < x_{\text{MIN}} \\ x_{\text{MAX}}, & x_{\text{MAX}} < x \end{cases} \quad (8)$$

$$\theta_{\text{new}} = \theta_{\text{old}} + \alpha \nabla_{\theta} L^{\text{clip}}(\theta) \quad (9)$$

$$L(\phi) = \mathbb{E}_{s \sim \pi_{\theta_t}} \left[\left(V_{\phi}(s_t) - V_t^{\text{target}} \right)^2 \right] \quad (10)$$